ELEC 378 Final Project of 2026 Spring

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Introduction

The final project gives you a "hands-on" opportunity to apply the techniques you've learned/will learn in machine learning! The project is intended to not only give you an opportunity to demonstrate your ability to apply the course material, but also to strongly encourage you to explore new concepts and techniques. This year, you will all work in groups to build one in-class model and one out-of-class model to classify butterfly and moth species from a given bank of pictures.

If you have any questions to this document or the Kaggle competition, feel free to reach out in Piazza!

Dataset

The dataset for this year's project consists of **12,594 labeled training images** and **1000 unlabeled test images** drawn from **100 butterfly and moth categories**. The training data are provided with class labels, while the test data are provided without labels and must be used for Kaggle submission. Each test image is identified by a unique `image_id`, and students must submit one predicted class label for every test image. All images are color butterfly or moth photographs prepared for a multiclass image classification task.

For grading, the **1000 test images are split into two parts**: approximately **50% public test data** and **50% private test data**. During the Kaggle competition, you will only see your score on the **public portion** of the test set, which is used for the live leaderboard. The **private portion** is hidden and will be used for the **final evaluation and ranking** after the competition closes. This means that strong public leaderboard performance does not guarantee strong final performance. If you overfit your model to the public leaderboard, your final ranking may drop when evaluated on the private test set. Keep this in mind when designing, tuning, and validating your models.

Getting Started

There is a skeleton code, skeleton.py, on the Kaggle Competition, to help the teams get started to load the datasets and check each image label pair.

Grading

Specifically, the grading on the final project is as follows:

- 20% - Kaggle leaderboard accuracy on the private test data. The TAs will observe the performance of Kaggle submissions and release a grading function based on the ranking and accuracy one week from the final deadline.
- 80% - The report quality as defined in the Report Section

Rules and Expectations

- You are not allowed to hard code the true labels of the images in the test set in any way.
- You must use and write in your report on at least TWO different models.
- At least one of your models must be one not introduced in class.
- At least one of your models must be non-neural network based.
- You are allowed to use machine learning toolboxes and deep learning packages.
- You are NOT allowed to use pre-trained models.
- You are NOT allowed to use outside data that isn't provided on our Kaggle competition.
- You are NOT allowed to copy another group's code.
- You are, however, allowed to reference external resources, provided you can explain the code and cite it in your report.

Report

Each team should submit one report for the final project. Your report should clearly document the full machine learning pipeline you developed for butterfly and moth species classification. A strong report does not only present the final model, but explains how your team explored the data, designed the pipeline, compared models, and justified the final submission.

Note that the percentage points aside each report section noted below sum to the 80% **weight of the report on your group's final project score; they are NOT x% of the** 80%**.**

In the report, include the following:

- **Team information (5%)** Include your team name shown on Kaggle, and each teammate's name, NetID, and contribution to the project.
- **Data exploration (5%)** Describe the dataset before modeling. Summarize what you observed about the images, classes, class balance, image sizes, visual variability, and any challenges in the dataset (for example, background clutter, pose variation, or visually similar species). Explain how you split the provided training data into training and validation sets, and justify that split.
- **Preprocessing and feature/design choices (20%)** Describe all preprocessing steps applied to the images before training. This may include resizing, normalization, cropping, color processing, augmentation, hand-crafted feature extraction, or other transformations. If you experimented with multiple preprocessing or feature choices, compare them and explain why your final choices were kept or discarded.
- **Models considered (10%)** You must discuss at least two different models, including the required in-class model and out-of-class model. For each model:
  - explain why you chose to try it,
  - describe the model architecture or algorithm,
  - state the key design choices and hyperparameters,
  - explain any course concepts used in the in-class model,
  - include the loss function and/or training objective when applicable.

  At least one model discussion must connect directly to course material, and at least one must explain a method not introduced / covered in depth in class.

- **Training procedure and tuning (10%)** Describe how each model was trained and tuned. Include details such as optimizer choice, learning rate, batch size, number of epochs, stopping criteria, regularization methods, and how hyperparameters were selected. If your team changed the pipeline over time, explain what was changed and why.
- **Quantitative comparison of models (10%)** Carefully compare the performance of your models. Do not only state which one performed best. Include quantitative results on the splits you used (for example, training and validation, and Kaggle public score if relevant). Discuss strengths and weaknesses of each model, and support your analysis with figures or tables.
- **Overfitting and generalization analysis (10%)** Discuss evidence of overfitting or underfitting in your experiments. Relate model complexity to dataset size and task difficulty. Explain what strategies you used to improve generalization, such as augmentation, regularization, early stopping, simplified models, or better validation procedures. If there is a gap between validation performance and Kaggle performance, analyze what that suggests.
- **Final submitted pipeline 5%** Describe the exact pipeline used for your final Kaggle submission from input image to predicted class. This should include preprocessing, model choice, training setup, and prediction procedure. Include at least one clearly labeled figure or diagram summarizing the full pipeline.
- **Results interpretation and conclusions 5%** Summarize your final results and explain why you believe the final method performed as it did. Discuss which parts of the pipeline mattered most to performance. Reflect on what you learned from the project, what surprised you, and what you would do differently if you had more time.
- **Code submission 20%** Submit your team's source code together with the report. This may be a GitHub link or a zip file with a clear README describing how to run the code. If you want us to reproduce your final model more easily, you are encouraged to also submit the final checkpoint(s).

GOOD LUCK HAVE FUN!

Figure 1: ELEC378 students doing the project